Assignment - due 17.09.2026

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Client Requirements Summary

The client has tasked us with configuring a multi-site enterprise network. This multi-site network consists of multiple geographic locations and a centralized data center. The client has specified that each site requires a specific amount of hosts and will need to be configured with the 173.50.32.0 address in mind.

For our client, they wished to configure 3 sites, each with different host requirements. Site 1 required 19 hosts, site 2 needed 24 and site 3 needed 50 hosts. There were a total of 6 routers to configure as well, these also had specific host requirements, ranging from 2 hosts to 5.

VLSM Process

Host Chart

256 128 64 32 16 8 4 2 --> chart used to determine how many bits we need to save

Steps

1. Given an address like this:
    - 173.50.32.0 / 19
    - The 19 represents our network bits, or the bits that dont change
    - Minus these from 32 to get the amount of host bits in our subnet
    - In this case 32-19 = 13, so we put 13 0s into our subnet.

- 11111111.11111111.11100000.00000000
- Resulting in our subnet

2. Use Host Chart to find the amount of bits we need to save
    - Given the fact we need 19 hosts, we consult the chart to find a number thats above our target.
    - In our host chart, it took us 5 bits to get to 32.
3. From the right end to the left, 'save' the amount of bits
    - Given 32, we need to save 5 bits in our subnet, the rest become 1s

- 11111111.11111111.11100000.00000000 becomes
    - 11111111.11111111.11111111.11100000

4. Convert this binary back to decimal and we have our subnet
   255.255.255.224
5. The increment of the range of our networks can be found by dictating the last 1 in our new subnet
   - In this example, it is found in the last octet: 11100000
   - this can be put on our normal binary chart to give us the range
   - The last one on our octet matches to 32.

- From our initial address:
  - 173.50.32.1-173.50.32.31
    - notice how its 31 and not 32, thats cause the next address is reserved for a different network (broadcast id).
    - 173.50.32.0 is reserved for our network id

6. make new / number for our new network by minusing the new host bits from our network bits.

VLSM Results

Initial IP = 173.50.32.0/21

50 Hosts - Data Center

Initial IP = 173.50.32.0/21
32 - 21 = 19

Initial subnet = 11111111.11111111.11111000.00000000

Save 6 bits

New subnet =
11111111.11111111.11111111.11000000
255.255.255.192

64 increment

173.50.32.0-173.50.32.63 (64)
Network ID = 173.50.32.0
Usable IPs: 173.50.32.1-173.50.32.63
Broadcast ID = 173.50.32.64

Final IP = 173.50.32.0/26

24 Hosts - Site 2

Initial IP = 173.50.32.64/26
32 - 26 = 6

Initial Subnet = 11111111.11111111.11111111.11000000

Save 5 bits

New subnet =
11111111.11111111.11111111.11100000
255.255.255.224

32 increment

Range =
173.50.32.64-173.50.32.95 (96)
/27

Final IP = 173.50.32.64/27

19 Hosts - Site 1

Initial IP = 173.50.32.96/27
32 - 27 = 5

Initial Subnet = 11111111.11111111.11111111.11100000

save 5 bits

New subnet = 11111111.11111111.11111111.11100000
255.255.255.224

32 increment

Range =
173.50.32.96-173.50.32.127 (128)

Final IP = 173.50.32.96/27

15 Hosts - Site 3

Initial IP = 173.50.32.128/27
32-27 = 5

Initial Subnet = 11111111.11111111.11111111.11100000
save 5 bits

New Subnet = 11111111.11111111.11111111.11100000
255.255.255.224
32 increment

Range = 173.50.32.128-173.50.32.159 (160)
173.50.32.129
173.50.32.130
173.50.32.131
173.50.32.132
173.50.32.133
173.50.32.134

5 Hosts - router3-6

initial ip = 173.50.32.160/27
32 - 27 = 5 host bits

initial subnet = 11111111.11111111.11111111.11100000

save 3 bits

new subnet = 11111111.11111111.11111111.11111000
255.255.255.248

increment by 8
Range = 173.50.32.160 - 173.50.32.167 (168 next)

Final network IP = 173.50.32.160/29

3 Hosts - router 2-4-5

initial ip = 173.50.32.168/29
32 - 29 = 3 host bits

initial subnet = 11111111.11111111.11111111.11111000

save 3 bits

new subnet = 11111111.11111111.11111111.11111000
255.255.255.248

increment by 8
Range = 173.50.32.168 - 173.50.32.175 (176 next)

Final network IP = 173.50.32.168/29

2 Hosts - router 1-2

initial IP = 173.50.32.176/29
32 - 29 = 3 host bits

initial subnet =11111111.11111111.11111111.11111000

save 2 bits
new subnet = 11111111.11111111.11111111.11111100
255.255.255.252
increment by 4
Range = 173.50.32.176 - 173.50.32.179 (180)
Final ip = 173.50.32.176/30

2 Hosts - router 1-3

Initial IP = 173.50.32.180/30
32 - 30 = 2 host bits

initial subnets = 11111111.11111111.11111111.11111100
save 2 bits
new subnet = 11111111.11111111.11111111.11111100
255.255.255.252

increment by 4
range = 173.50.32.180-173.50.32.183 (184)

Final IP = 173.50.32.180/30

2 Hosts - router 2-3

Inital IP = 173.50.32.184/30

32 - 30 = 2 host bits

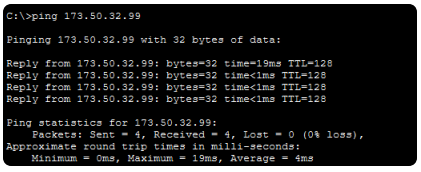
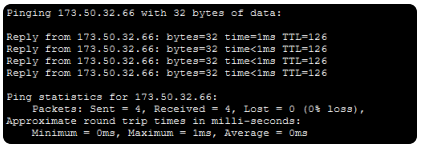
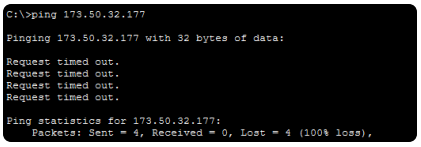
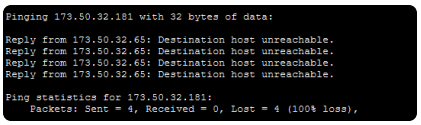
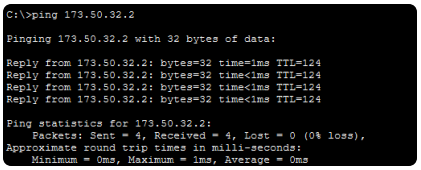
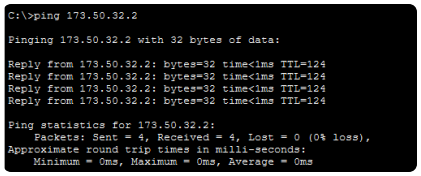
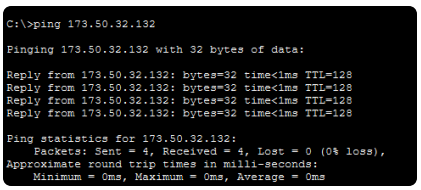
initial subnets = 11111111.11111111.11111111.11111100
save 2 bits
new subnet = 11111111.11111111.11111111.11111100
255.255.255.252

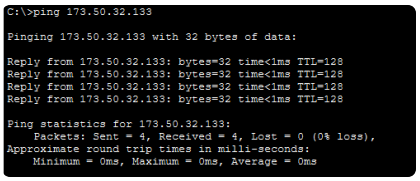
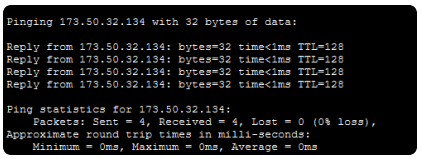
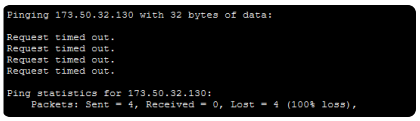
increment by 4
range = 173.50.32.184-173.50.32.187 (188)

Final IP = 173.50.32.184/30

Configuration Summary

Inter-Network Connectivity Verification

Source	Source IP Address	Destination	Destination IP Address	Pass/Fail
PC0	173.50.32.98	PC1	173.50.32.99	Pass 
PC0	173.50.32.98	PC8	173.50.32.66	Pass 
PC0	173.50.32.98	Router 1 Gig0/0	173.50.32.177	Fail 
PC8	173.50.32.66	Router 1 Gig0/2	173.50.32.181	Fail 
PC0	173.50.32.98	Server1	173.50.32.2	Pass 
PC9	173.50.32.67	Server0	173.50.32.2	Pass 
PC2	173.50.32.129	PC5	173.50.32.132	Pass 

Source	Source IP Address	Destination	Destination IP Address	Pass/Fail
PC3	173.50.32.130	PC6	173.50.32.133	Pass 
PC4	173.50.32.131	PC7	173.50.32.134	Pass 
PC2	173.50.32.129	PC3	173.50.32.130	Fail 

Task 8 Questions

1. The organization plans to expand by adding two new sites, what physical topology options, changes, or additions could be considered for the network design? Please answer with reference to the provided network topology and include a diagram for your suggested network topology.
2. Explain why VLSM is a better choice for this network instead of fixed size subnets
   VLSM is a better choice as it eliminates the need to guess the subnet for this network, it also reduces the amount of wasted IPs generated. It also enables scalability, which is highly beneficial as this organization is planning to expand their network.
3. What is the role of the default gateway (configured on a PC) during communication?
   The default gateway acts as the forwarding host for nodes in a network. It is responsible for forwarding the packets sent to it to a designated network.
4. What is the relationship between virtualization and AWS in cloud computing?

Virtualization is the process of using software to split a physical machine into numerous virtual machines. This method is utilized in AWS to split massive pieces of physical hardware into smaller machines for customers to use.

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Server0	-	173.50.32.2	255.255.255.192	173.50.32.1
Server1	-	173.50.32.3	255.255.255.192	173.50.32.1

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router6	G0/2	173.50.32.1	255.255.255.192	-
Router6	G0/0	173.50.32.162	255.255.255.248	-
PC2	VLAN10	173.50.32.129	255.255.255.224	0.0.0.0
PC3	VLAN20	173.50.32.130	255.255.255.224	0.0.0.0
PC4	VLAN30	173.50.32.131	255.255.255.224	0.0.0.0
PC5	VLAN10	173.50.32.132	255.255.255.224	0.0.0.0
PC6	VLAN20	173.50.32.133	255.255.255.224	0.0.0.0
PC7	VLAN30	173.50.32.134	255.255.255.224	0.0.0.0
Router3	G0/0	173.50.32.161	255.255.255.248	-
Router3	G0/1	173.50.32.185	255.255.255.252	-
Router3	G0/2	173.50.32.182	255.255.255.252	-
Router1	G0/2	173.50.32.181	255.255.255.252	-
Router1	G0/0	173.50.32.177	255.255.255.252	-
Router2	G0/0	173.50.32.178	255.255.255.252	-
Router2	G0/1	173.50.32.186	255.255.255.252	-
Router2	G0/2	173.50.32.169	255.255.255.248	-
Router4	G0/0	173.50.32.170	255.255.255.248	-
Router4	G0/1	173.50.32.97	255.255.255.224	-
PC0	Fa0	173.50.32.98	255.255.255.224	173.50.32.97
PC1	Fa0	173.50.32.99	255.255.255.224	173.50.32.97
Router5	G0/0	173.50.32.171	255.255.255.248	-
Router5	G0/1	173.50.32.65	255.255.255.224	-
PC8	Fa0	173.50.32.66	255.255.255.224	173.50.32.65
PC9	Fa0	173.50.32.67	255.255.255.224	173.50.32.65